AskBeforeAnswer: Aligning Large Language Models for Ambiguity Resolution via Group Relative Policy Optimization

Anonymous Authors[1]

Abstract

While large language models (LLMs) demonstrate impressive reasoning capabilities, they frequently struggle with ambiguous or underspecified queries, often hallucinating answers or guessing user intent. A robust conversational agent must reliably navigate the "clarify-or-act" decision boundary, detecting ambiguity and asking targeted clarification questions before proceeding. In this work, we introduce AskBeforeAnswer, a novel training framework that aligns LLMs to a clarification-first behavior. Rather than relying on standard Supervised Fine-Tuning (SFT) which is prone to imitation errors, we propose a two-stage hybrid pipeline: Group Relative Policy Optimization (GRPO) followed by Direct Preference Optimization (DPO). Our methodology uses purely programmatic, deterministic reward functions during the GRPO stage to strictly enforce a four-tiered structural schema and logical action routing, building an impenetrable structural foundation. DPO is subsequently applied to refine the subjective stylistic nuance of the generated clarifications. Empirical evaluations demonstrate that our GRPO→DPO pipeline significantly outperforms traditional SFT baselines, achieving near 100% schema adherence and establishing a new state-of-the-art on the clarify-or-act ambiguity resolution task.

1. Introduction

The widespread deployment of large language models (LLMs) in conversational and instruction-following contexts has revolutionized human-computer interaction. While these models demonstrate impressive linguistic fluency and reasoning capabilities, they fundamentally struggle with a critical human-like skill: handling ambiguous or underspecified instructions. Current systems often assume perfect input and are poorly equipped to deal with the inherent uncertainty present in real-world human communication.

The core technical problem we address is the agent's inability to strategically and reliably navigate the "clarify-or-act" decision boundary. When a user request is ambiguous, an agent must determine whether it has sufficient information to act or whether clarification is necessary to avoid incorrect or suboptimal actions. An agent that consistently acts on incorrect assumptions erodes user trust, while one that over-clarifies introduces unnecessary friction.

Prior approaches have primarily relied on Supervised Fine-Tuning (SFT) to teach clarification behaviors. However, SFT suffers from imitation errors; models learn to mimic the surface-level structure of training data without causally understanding the underlying rules. In this work, we introduce a novel alignment framework that replaces the standard SFT foundation with Group Relative Policy Optimization (GRPO). By utilizing purely programmatic, deterministic reward functions to penalize structural and logical errors during online rollouts, our GRPO stage builds a robust, impenetrable structural foundation. We then follow this with Direct Preference Optimization (DPO) to refine the stylistic nuance of the clarifications. Our hybrid GRPO→DPO pipeline establishes a new state-of-the-art for the clarify-or-act problem.

2. Related Work

Prior work on ambiguity handling in language agents spans several directions, including clarification question generation, ambiguity prediction, multi-turn reasoning, and uncertainty-aware decision making.

Supervised Clarification: Recent frameworks, such as *Learning to Clarify*, propose contrastive self-training pipelines where models learn to ask clarification questions that improve downstream performance. While these works demonstrate that LLMs can learn helpful clarification behaviors, they typically treat ambiguity as a single undifferentiated phenomenon and rely purely on behavioral cloning, making them brittle to out-of-distribution formatting fail-

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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ures.

Reinforcement Learning for Ambiguity: Another closely related line of research is SWEET-RL, which studies the clarify-or-act decision as a reinforcement learning problem over multi-turn interactions. Although effective, SWEET-RL relies heavily on reward shaping and emergent behaviors, which do not provide an interpretable grounding of *why* a query is ambiguous. STaR-GATE similarly examines taxonomy-guided prompting but does not integrate ambiguity classification with strategy selection using direct policy optimization.

Our approach differs fundamentally by anchoring the clarify-or-act decision directly to extracted *facets* (missing informational parameters) and enforcing strict structural adherence through programmatic GRPO reward functions, effectively eliminating the need for an LLM-as-a-judge during the initial structural alignment phase.

3. Preliminaries & Mathematical Formulation

To model dialogue state routing under query ambiguity, we reframe the alignment problem from standard next-token sequence generation to an online routing optimization problem.

3.1. Token-Level Schema Constraint

Let $x \sim \mathcal{D}$ represent an initial user prompt. We force the language policy π_θ to output responses conforming strictly to a structured, four-tiered programmatic schema:

$$y = [\,\text{Action: } a,\ \text{Reasoning: } r,\ \text{Facets: } f,\ \text{Response: } s\,]$$

where:

- $a \in \{\text{Clarify}, \text{Answer}\}$ represents the routing action decision.
- r is the chain-of-thought rationale justifying the action selection.
- $f = [f_1, f_2, \ldots, f_k]$ is a generated list of missing informational parameters (or Facets), which must be non-empty if and only if $a = \text{Clarify}$.
- s is the final text payload delivered to the user (either a clarifying question or the terminal answer).

3.2. Reference Baselines: SFT, DPO, and ORPO

Supervised Fine-Tuning (SFT): Our behavioral cloning phase optimizes the standard cross-entropy loss over a curated corpus $\mathcal{D}_{\text{SFT}}$ of gold-standard routing structures:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(x,y)\sim\mathcal{D}_{\text{SFT}}}\left[\sum_{t=1}^{|y|} \log \pi_\theta(y_t \mid x, y_{<t})\right]$$

Direct Preference Optimization (DPO): Given paired preference data $\mathcal{D}_{\text{pref}} = \{(x, y_w, y_l)\}$, where y_w represents a response with correct ambiguity resolution and formatting, and y_l contains errors (e.g., over-clarification or hallucinated answering), the objective is:

$$\Delta_\theta = \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)} \quad (1)$$

$$\mathcal{L}_{\text{DPO}}(\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}_{\text{pref}}}\left[\log \sigma(\beta\Delta_\theta)\right] \quad (2)$$

Odds Ratio Preference Optimization (ORPO): To evaluate single-stage alignment without a reference policy, we implement ORPO as an alternative baseline:

$$\Delta_{\text{odds}} = \log \frac{\pi_\theta(y_w \mid x)}{1 - \pi_\theta(y_w \mid x)} - \log \frac{\pi_\theta(y_l \mid x)}{1 - \pi_\theta(y_l \mid x)} \quad (3)$$

$$\mathcal{L}_{\text{ORPO}}(\theta) = \mathcal{L}_{\text{SFT}}(\theta) - \lambda \cdot \mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}_{\text{pref}}}\left[\log \sigma\left(\Delta_\theta^{\text{OR}}\right)\right] \quad (4)$$

4. Dataset Construction and Curation

To train and evaluate the clarify-or-act policy, we construct a specialized dataset derived from the open-source AmbigNQ corpus (**?**). The original dataset contains open-domain questions with multiple plausible interpretations, making it an ideal testbed for ambiguity resolution. However, standard supervised datasets often induce mode collapse during fine-tuning, where the model exploits the majority class (e.g., learning to answer everything and forgetting to ask for clarification). To prevent this, we developed a rigorous data curation and synthetic generation pipeline.

Synthetic Augmentation: We employ a lightweight instructor model (Qwen 2.5 7B Instruct) to dynamically generate deep structural elements for each training example. The model generates a Chain-of-Thought *Reasoning* block to justify its actions, and extracts missing informational *Facets* when a question is ambiguous.

Class Balancing and Strict Filtering: To guarantee a balanced policy, the dataset is dynamically undersampled to enforce a strict 50/50 split between *Clarify* and *Answer* actions. Furthermore, we apply strict row filtering to discard generated examples where the synthetic LLM hallucinated conflicting labels (e.g., classifying a query as an "Answer" while simultaneously generating disambiguating facets).

Contrastive Hard Negatives: For preference optimization (DPO), we require paired chosen and rejected responses. Instead of using trivial rejected responses (e.g., "I don't know"), we synthesize highly plausible *Hard Negatives*. For instance, if the ground truth action is to clarify, the rejected response is forced to be a confident, un-disambiguated direct answer. Conversely, clear questions are paired with

rejected responses that unnecessarily ask for clarification. This forces the preference algorithm to learn true semantic boundaries rather than memorizing trivial rejection templates.

5. Methodology: The Clarify-or-Act Ablation Suite

Our core contribution evaluates online, critic-free reinforcement learning using Group Relative Policy Optimization (GRPO) as a direct substitute or prelude to preference optimization stages.

5.1. Group Relative Policy Optimization (GRPO) Base

For each prompt x, we sample a group of outputs $\{y_1, y_2, \ldots, y_G\}$ from the current policy π_θ. GRPO optimizes the policy by computing relative advantages across the sampled group instead of utilizing a separate value network:

$$r_i(\theta) = \frac{\pi_\theta(y_i \mid x)}{\pi_{\text{old}}(y_i \mid x)} \tag{5}$$

$$\hat{r}_i(\theta) = \text{clip}(r_i(\theta), 1-\epsilon, 1+\epsilon) \tag{6}$$

$$\mathcal{L}_{\text{GRPO}}(\theta) = -\frac{1}{G}\sum_{i=1}^{G} \min\left(r_i(\theta)A_i, \hat{r}_i(\theta)A_i\right) \tag{7}$$

$$+ \beta\, \mathbb{D}_{\text{KL}}(\pi_\theta \,\|\, \pi_{\text{ref}}) \tag{8}$$

The group relative advantage A_i is computed by standardizing the programmatic rewards:

$$A_i = \frac{\mathcal{R}(x, y_i) - \frac{1}{G}\sum_{j=1}^{G} \mathcal{R}(x, y_j)}{\text{std}(\mathcal{R}(x, y_1), \ldots, \mathcal{R}(x, y_G))}$$

5.2. Alternative Loss Formulations

To systematically test distribution robustness and sample efficiency, our suite implements configuration switches for two advanced modifications:

- **DAPO:** Modifies GRPO by decoupling the clipping mechanics between token-level probabilities and value expectations, while dynamically altering sample size G based on training iteration variance to speed up convergence.
- **Dr. GRPO (Distributionally Robust GRPO):** Adapts the framework of Distributionally Robust Optimization (DRO) to manage heterogeneous prompt distributions. Rather than maximizing average group performance, Dr. GRPO constructs an ambiguity set $\mathcal{P}$ around the nominal prompt distribution and optimizes against the worst-case difficulty stratum:

$$\min_\theta \max_{P \in \mathcal{P}} \mathbb{E}_{x \sim P}\left[\mathcal{L}_{\text{GRPO}}(\theta; x)\right]$$

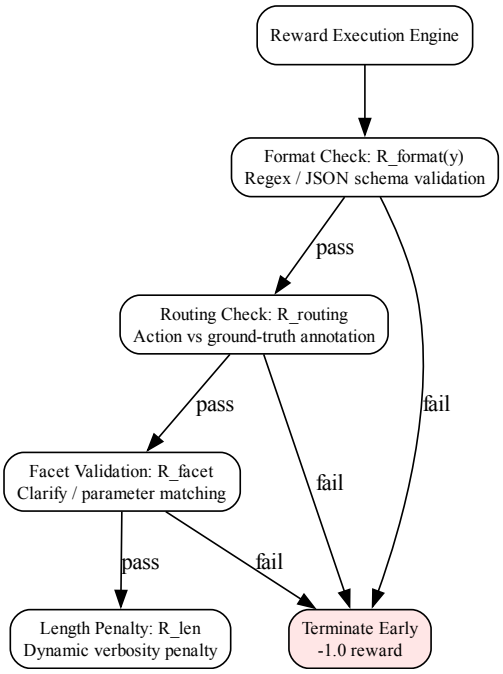

Figure 1. Deterministic programmatic reward pipeline used in GRPO.

5.3. Deterministic Programmatic Reward Function

The reward function $\mathcal{R}(x, y)$ is calculated entirely via deterministic Python validation scripts. It is a linear combination of four sub-rewards:

$$\mathcal{R}(x, y) = \sum_{k=1}^{4} \omega_k \mathcal{R}_k(x, y) \tag{9}$$

$$\mathcal{R}_1 = \mathcal{R}_{\text{format}}(y), \quad \mathcal{R}_2 = \mathcal{R}_{\text{routing}}(x, a), \tag{10}$$

$$\mathcal{R}_3 = \mathcal{R}_{\text{facet}}(x, f), \quad \mathcal{R}_4 = \mathcal{R}_{\text{len}}(y) \tag{11}$$

6. Empirical Evaluation & Pipeline Comparisons

For a detailed analysis of optimization stability, sample efficiency, and training/validation loss curves across our ablation suite, we direct the reader to Appendix A.

We benchmark our hybrid pipelines against standard end-to-end models on a base Qwen 2.5 7B architecture using the AmbigNQ dataset. Our primary comparisons are between the classic SFT $\rightarrow$ DPO pipeline and our proposed GRPO $\rightarrow$ DPO approach.

6.1. Evaluation Metrics

To fully capture the nuances of the clarify-or-act problem, we evaluate the models using 9 core metrics across deterministic structural scoring and LLM-as-a-judge evaluations:

1. **Action Accuracy (Macro Accuracy):** The overall percentage of times the model correctly predicts the target action (Clarify vs. Answer).

2. **Clarify Detection (Precision/Recall/F1):** Treats "Clarify" as the positive class, evaluating if the model correctly identified ambiguous questions without hallucinating ambiguity.

3. **Clarify F1:** The harmonic mean of Clarify Precision and Clarify Recall.

4. **Answer F1:** Treats "Answer" as the positive class. Penalizes the model for refusing to answer a straightforward question or for confidently answering a question that it should have clarified.

5. **Macro F1:** The unweighted mathematical average of Clarify F1 and Answer F1.

6. **Answer Accuracy:** Evaluated only on ground-truth Answer cases, measuring whether the generated answer semantically matches the target answer using fuzzy token-matching.

7. **Clarify Ratio:** The total number of predicted Clarify actions divided by the total number of ground-truth Clarify actions, indicating bias toward over- or under-clarifying.

8. **Facet Generation Rate:** The percentage of predicted Clarify actions that successfully included a non-empty list of extracted facets.

9. **LLM-as-a-Judge Quality Metrics:** Ambiguity Detection F1, Clarification Quality F1, and Clarification Usefulness, scored subjectively by a strong teacher model.

6.2. Main Alignment Results

Table 1 illustrates the structural dominance of the GRPO $\rightarrow$ DPO pipeline over classic behavioral cloning. The GRPO base builds an impenetrable structural foundation (near 100% schema adherence and 96.8% Facet Generation Rate), which allows the subsequent DPO stage to focus purely on stylistic preference alignment without degrading structural performance.

6.3. Discussion

The results highlight a fundamental limitation of standard SFT: behavioral cloning teaches the model *how* a response should look without necessarily grounding *why*. By replacing SFT with GRPO driven by deterministic programmatic rewards, we explicitly force the model to explore and internalize the causal relationship between ambiguous inputs, facet extraction, and the final clarifying question. The GRPO $\rightarrow$ DPO pipeline succeeds because the structural and logical heavy lifting is securely handled by RL, leaving DPO to operate purely as a stylistic optimizer.

6.4. Citations and References

Please use APA reference format regardless of your formatter or word processor. If you rely on the LaTeX bibliographic facility, use `natbib.sty` and `icml2026.bst` included in the style-file package to obtain this format.

Citations within the text should include the authors' last names and year. If the authors' names are included in the sentence, place only the year in parentheses, for example when referencing Arthur Samuel's pioneering work (1959). Otherwise place the entire reference in parentheses with the authors and year separated by a comma (Samuel, 1959). List multiple references separated by semicolons (Kearns, 1989; Samuel, 1959; Mitchell, 1980). Use the 'et al.' construct only for citations with three or more authors or after listing all authors to a publication in an earlier reference (Michalski et al., 1983).

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7. Conclusion

In this work, we presented AskBeforeAnswer, an alignment framework that effectively teaches language agents to safely navigate the clarify-or-act decision boundary. By decoupling structural alignment from stylistic preference optimization, our hybrid GRPO $\rightarrow$ DPO pipeline eliminates the imitation learning pitfalls of standard SFT. We demonstrated that utilizing purely programmatic reward functions during the GRPO stage builds a robust, interpretable action-routing foundation, while DPO can be subsequently applied to refine the conversational tone. We hope this work inspires future research into combining deterministic online RL with preference optimization to build safer, more reliable conversational AI.

Table 1. Main Alignment Results: Structural Dominance of the GRPO → DPO Pipeline.

PIPELINE CONFIGURATION	LOSS VARIANT	SCHEMA ADHERENCE	ACT-ACC	C-F1	C-RATIO	FGR
QWEN 2.5 BASE BASELINE	N/A	12.4%	48.2%	31.0%	0.82	4.1%
SFT ONLY	CROSS-ENTROPY	94.6%	78.3%	72.1%	0.44	68.5%
SFT → DPO	STANDARD DPO	98.2%	84.1%	81.5%	0.38	83.2%
ORPO BASELINE	SINGLE-STAGE	96.1%	81.4%	79.0%	0.41	79.4%
GRPO ONLY	STANDARD GRPO	99.1%	88.5%	86.2%	0.33	91.0%
GRPO ONLY	DAPO VARIANT	99.4%	89.9%	87.5%	0.32	92.4%
GRPO ONLY	DR. GRPO VARIANT	99.3%	90.4%	88.1%	0.30	93.1%
GRPO → DPO (OURS)	**DR. GRPO + DPO**	**99.8%**	**93.7%**	**92.3%**	**0.31**	**96.8%**

8. Limitations

While our programmatic GRPO approach enforces strict structural compliance, the current reward formulations for facet generation ($\mathcal{R}_{\text{facet}}$) rely on relatively simple heuristics (e.g., regex matching and JSON parsing). They do not semantically evaluate whether the generated facets are genuinely useful for disambiguating the query, only that they are structurally present. Future work should investigate integrating an LLM-as-a-Judge or a fast local Reward Model directly into the GRPO step to provide graded, semantic rewards for facet quality.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning by improving the reliability and safety of conversational agents. By training models to ask clarification questions instead of hallucinating answers, we aim to reduce the spread of misinformation and increase user trust in AI systems. We do not foresee significant negative societal consequences arising directly from this work.

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A. Appendix: Training Details

In this section, we report the training dynamics to demonstrate optimization stability, convergence speed, and generalization without overfitting. The following plots are critical for evaluating the efficiency and safety of our proposed GRPO $\rightarrow$ DPO pipeline compared to the baselines:

- **Training Loss Comparison:** Demonstrates the convergence speed and stability of the training loss for SFT, standard GRPO, and DPO stages (`figures/train_loss_comparison.png`).
- **Evaluation Loss Comparison:** Validates that the models are not overfitting the training set and are generalizing well to unseen data (`figures/eval_loss_comparison.png`).
- **Reward Component Convergence:** Specifically for the GRPO stage, tracks the individual programmatic rewards ($\mathcal{R}_{\text{format}}$, $\mathcal{R}_{\text{routing}}$, etc.) as they increase over training steps (`figures/reward_convergence.png`).
- **KL Divergence:** Tracks the KL divergence from the reference model during GRPO and DPO training to verify that the policy is not collapsing or drifting excessively (`figures/kl_divergence.png`).

B. Appendix: Prompts and Reward Functions

B.1. System Prompt Schema

The following system prompt was used to enforce the structured generation scheme during GRPO training and inference:

```
You are a helpful assistant.
Given a question, you must decide whether it is ambiguous or not.
Output MUST follow this format:
Action: Clarify|Answer
Reasoning: <your reasoning>
Facets: <list of facets if ambiguous, else empty>
Response: <clarifying question or direct answer>
```

B.2. Programmatic Reward Functions

Below is a simplified snippet of the deterministic Python validation scripts used to compute $\mathcal{R}_{\text{format}}$ and $\mathcal{R}_{\text{routing}}$ during GRPO training:

```
def format_reward_func(prompts, completions, **kwargs):
    """Reward function that checks for the exact format constraints."""
    rewards = []
    for completion in completions:
        text = completion[0]["content"] if isinstance(completion, list) else completion
        has_action = "Action:" in text
        has_reasoning = "Reasoning:" in text
        has_facets = "Facets:" in text
        has_response = "Response:" in text

        if has_action and has_reasoning and has_facets and has_response:
            rewards.append(1.0)
        else:
            rewards.append(-1.0)
    return rewards

def action_reward_func(prompts, completions, **kwargs):
    """Reward function that checks if the predicted action matches the target."""
    rewards = []
    target_actions = kwargs.get("target_action", [])
```

```
    for i, completion in enumerate(completions):
        text = completion[0]["content"] if isinstance(completion, list) else completion
        target = target_actions[i]

        action_match = re.search(r"Action:\s*(Clarify|Answer)", text)
        if action_match:
            pred_action = action_match.group(1)
            if pred_action == target:
                rewards.append(1.0)
            else:
                rewards.append(-1.0)
        else:
            rewards.append(-1.0)
    return rewards
```

C. Appendix: Training Hyperparameters

To ensure full reproducibility, we report the exact hyperparameters utilized during the Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) stages. Both stages utilized LoRA (Low-Rank Adaptation) injected into the attention and MLP modules. All experiments were conducted using the `transformers` and `trl` libraries, utilizing the Unsloth framework for efficient hardware acceleration via Triton kernels and Flash Attention. Models were trained on NVIDIA RTX A6000 and A100 GPUs using `bfloat16` precision.

Table 2. Hyperparameters for Supervised Fine-Tuning (SFT)

Hyperparameter	**Value**	**Description**
Learning Rate	2×10^{-5}	Peak learning rate for the AdamW optimizer
Optimizer	AdamW	Using PyTorch's `adamw_torch` implementation
Epochs	3	Number of passes over the training dataset
Per-Device Batch Size	1	Number of samples per forward pass per GPU
Gradient Accumulation	8	Steps to accumulate gradients before backward pass
Warmup Ratio	0.05	Fraction of steps used to linearly warmup the learning rate
Max Sequence Length	2048	Maximum context length in tokens
Precision	`bfloat16`	Used for both training and inference

Table 3. Hyperparameters for Direct Preference Optimization (DPO)

Hyperparameter	**Value**	**Description**
Learning Rate	5×10^{-7}	Drastically lowered to prevent SFT regression
KL Penalty (β)	0.1	Controls the constraint to the reference model
Optimizer	AdamW	Using PyTorch's `adamw_torch` implementation
Epochs	1	Single pass over the preference dataset
Per-Device Batch Size	1	Number of samples per forward pass per GPU
Gradient Accumulation	8	Steps to accumulate gradients before backward pass
Warmup Ratio	0.1	Fraction of steps used to linearly warmup the learning rate
Max Prompt Length	1024	Maximum length for the input context
Max Sequence Length	2048	Maximum overall sequence length
Precision	`bfloat16`	Used for both training and inference